

CARLA PENAFIEL MESSMER

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Education

2026–2027	M.S. in Data Science , Columbia University DSI Scholar , Data Science Institute – “Mapping Children’s Vulnerability”	New York, NY, USA (<i>Expected Dec. 2027</i>)
2020–2023	M.A. in Economics , University of Chile Class Rank: 2nd of 21 Thesis: “Gender conditions, labor market and child abuse in Chile” (<i>Honors</i>)	Santiago, Chile
2017–2023	B.S. in Industrial Engineering , University of Chile Class Rank: 1st of 174 (<i>Honors</i>)	Santiago, Chile

Professional Experience

2021–2026	Bain & Company Lead Data Scientist	Santiago, Chile 2025–2026
	- Led data science workstreams across pricing, optimization, fraud detection, and predictive analytics for clients across Latin America, the U.S., and the Caribbean. - Built a marketing mix model for an Argentine wine company and optimization models for mining operations in Chile, including truck fleet optimization. - Developed price–demand elasticity models and pricing recommenders, forecasting the impact of pricing decisions on demand, revenue, and margin.	
	Senior Data Scientist	2023–2025
	- Built price–demand elasticity models and scenario-based pricing recommenders across chemicals, gas, and insurance, forecasting the impact of pricing decisions on demand, revenue, and margin. - Developed fraud-detection and early-warning solutions for banking, insurance, and utilities using LightGBM, NLP/BETO embeddings, anomaly detection, clustering, and supervised learning. - Built an insurance pricing optimization framework using LightGBM and Pyomo to maximize portfolio margin subject to renewal-rate constraints.	
	Data Scientist	2021–2023
	- Developed pricing, demand-elasticity, forecasting, and marketing-mix models for clients across Chile, Brazil, Mexico, and Argentina. - Developed optimization solutions for gas operations, improving operational efficiency and resource allocation.	

Research Experience

2020–2021	J-PAL Latin America and the Caribbean (J-PAL LAC) Research Assistant – Supported the design, implementation, and analysis of a randomized controlled trial on information frictions in mortgage refinancing. Prepared and processed experimental data, designed the testing statement used in the experiment, and contributed to the implementation and analysis of the RCT. The project resulted in <i>Frictions in Mortgage Refinancing: Evidence from Chile</i> .	Santiago, Chile
2021–2022	Inter-American Development Bank (IDB) Research Consultant – Conducted quantitative research on gender attitudes and perceptions among PhD holders in male-dominated fields, contributing to a co-authored research study.	Santiago, Chile

Teaching Experience

Graduate: Graduate Workshop on Applied Economics, MAGCEA (2023–2025); Graduate Economics, MBA (2021–2024).
Undergraduate: Statistics (2019–2021), Economics (2020), Macroeconomics (2019), University of Chile.

Awards & Leadership

Honor Student for four consecutive years (2019–2022), Faculty of Physical and Mathematical Sciences, University of Chile; **Women’s Scholarship in Economics and Science** (2020–2022); **Honor Student**, M.A. in Economics (2023), University of Chile; **Founder**, Mentoring Program for M.A. in Economics students (2021–2023), University of Chile.

Technical Skills & Languages

Programming: Python (pandas, NumPy, scikit-learn, statsmodels, XGBoost, LightGBM), R, Stata, SQL; **ML & Analytics:** NLP, agentic AI, anomaly detection, clustering, supervised learning, forecasting, predictive analytics, pricing optimization, Pyomo; **Tools:** Git/GitHub, Power BI, Streamlit; **Languages:** Spanish (native), Portuguese (advanced)

Publications

Co-author, chapter “Vida cotidiana en Pandemia por COVID-19 en Chile” in *Pandemia y Crisis*, 2022; manuscript in preparation for journal submission: “Gender conditions, labor market and child abuse in Chile”. with P. Jervis.